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Equidistribution of generalized Dedekind sums and exponential sums ☆

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ABSTRACT

For the generalized Dedekind sums $s_{ij}(p,q)$ defined in association with the x^iy^j-coefficient of the Todd power series of the lattice cone in $\mathbb{R}^2$ generated by $(1,0)$ and (p,q), we associate an exponential sum. We obtain this exponential sum using the cocycle property of the Todd series of 2d cones and the nonsingular cone decomposition along with the continued fraction of q/p. Its Weil bound is given for the modulus q applying the purity theorem of the cohomology of the related $\bar{\mathbb{Q}}_\ell$-sheaf due to Denef and Loeser. The Weil type bound of Denef and Loeser fulfills the Weyl equidistribution criterion for $R(i,j)q^{i+j-2}s_{ij}(p,q)$. As a special case, we recover the equidistribution result of the classical Dedekind sums multiplied by 12 not using the modular weight of the Dedekind $\eta(\tau)$.

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1. Introduction

Classical Dedekind sums $s(p,q)$ are defined for relatively prime integers p,q by

$$s(p,q)=\sum_{k=1}^{q}\left(\left(\frac{k}{q}\right)\right)\left(\left(\frac{kp}{q}\right)\right)$$

where $((x))$ denotes the value of the 1st periodic Bernoulli function at x:

$$((x))=\bar{B}_1(x):=\begin{cases}x-[x]-\frac{1}{2} & \text{for } x\notin\mathbb{Z},\\ 0 & \text{for } x\in\mathbb{Z}.\end{cases}$$

This appears important in describing the change of the Dedekind eta function

$$\eta(\tau)=e^{2\pi i\tau/24}\prod_{n=1}^{\infty}\left(1-e^{2\pi in\tau}\right),\quad \tau\in\mathfrak{h},$$

under modular transformations. $\eta(\tau)$ is a 24th root of the modular discriminant

$$\Delta(\tau)=(12\pi)^{12}\eta^{24}(\tau)$$

up to some constant.

Due to the modularity after 24th power, under modular transformation $\tau\mapsto A\tau$, for $A=\left(\begin{smallmatrix}a & b\\ c & d\end{smallmatrix}\right)\in \mathrm{SL}_2(\mathbb{Z})$, its logarithm satisfies the following formula due to Dedekind:

$$\log\eta\left(\frac{a\tau+b}{c\tau+d}\right)=\log\eta(\tau)+\frac{1}{4}\log\left(-(c\tau+d)^2\right)+\pi i\phi(A)\tag{1}$$

where $\phi(-)$ is the Rademacher ϕ-function defined as a function of $\mathrm{SL}_2(\mathbb{Z})$

$$\phi(A) := \begin{cases} \frac{a+d}{12c} - \operatorname{sign} c \cdot s(a,c) & \text{for } c \neq 0, \\ \frac{b}{12d} & \text{for } c = 0, \end{cases} \tag{2}$$

valued in $\frac{1}{12}\mathbb{Z}$. Rademacher's ϕ-function is valued in $\frac{1}{12}\mathbb{Z}$ and many interesting properties of Dedekind sums arise in study of ϕ (Rademacher's ϕ-function appeared in [30] is 12 times of (2) thus valued in $\mathbb{Z}$).

Besides the above modular transformation property, Dedekind sums and their generalization appear in some expressions of the special values of some L or zeta functions at nonpositive integers. Probably the most famous is the theorem of Meyer on class number formula in [26]. Siegel and Shintani obtained an expression of special values at nonpositive integers (cf. [34,35,33]). In particular, Shintani's expression lies in the same line of thought. One has expression of zeta values using some generalization of Dedekind sums [33,13]. Using this, one can have painless proof of the rationality of the special values at nonpositive integers.

We are interested in the distribution of the Dedekind sums and their generalization. In [18,20], we observed that the random distribution of Dedekind sums is closely related to the integrality of the special values of partial zeta functions. Originally, Rademacher and Grosswald raised a question on the density of the values of Dedekind sums in p. 28 of [30]. There is a long list of publications on this direction. Hickerson proved that the points $(p/q, s(p,q))$ are dense in $\mathbb{R}^2$ [14]. Much later, big progress was made independently by Myerson and Vardi. Vardi showed that for any positive real number r, the set $\{\langle rs(p,q)\rangle \mid 0 < p < q,\ (p,q) = 1\}$ is equidistributed on the interval $[0,1)$ by relating Kloosterman sums to Dedekind sums [39]. In [28], Myerson applied similar method to show that for a given nonzero real r the graph of the function $p/q \mapsto rs(p,q)$ (modulo 1) is equidistributed in the unit square.

Let us recall briefly the idea of Vardi and Myerson. As ϕ is valued in $\frac{1}{12}\mathbb{Z}$, for the case $r \in 12\mathbb{Z}$, one can relate the Kloosterman sums to the Dedekind sums multiplied by 12 (see Theorem 2.1 of this article). There is a well-known bound of Kloosterman sums for varying modulus due to Weil [40], which is crucial step in showing the Weyl equidistribution criterion for the fractional part of Dedekind sums. For $r \notin 12\mathbb{Z}$, a generalization of Kloosterman sums due to Selberg are associated to the multiplier system arising from Dedekind's η-function [32]. The strong equidistribution is beyond our interest and our discussion focuses more on the occurrence of the factor $\frac{1}{12}$ and its generalization.

This will be reviewed more precisely in Section 2 of this article for our own purpose.

In this article, we note that many interesting properties of Dedekind sums are consequence of cocycle condition. As a cochain over $(\mathrm{P})\,\mathrm{SL}_2(\mathbb{Z})$, the coboundary of the Rademacher ϕ-function is identified with either the area or the signature 2-cocycle of torus fibration over a pair of pants (cf. [4,15,21,27]). Since these 2-cocycles have simple geometric interpretations, one obtains identities involving Dedekind sums (cf. [3,4,15,21, 31]). It is a folklore that all known properties are consequences of the cocycle condition of

this sort (e.g. [37,36,31]). To name one of them, one obtains the celebrated reciprocity law for the Dedekind sums by swapping the two arguments. Also from the (finite length) continued fraction of a rational number one can describe it using the terms of the continued fraction (cf. [5,14,22]). Of course, a brute-force computation using the explicit expression recovers the reciprocity law again, but the explicit form itself is already the consequence of the cocycle condition. It is worth to note that Fukuhara identified Dedekind symbols in terms of the reciprocity function [11,12]. From the point of view of cocycle property, after small corrections Dedekind symbols are part of 1-cocycle and their reciprocity is the coboundary operation followed by specialization at a cusp. The cohomological understanding is given by Manin through [24,23,25] for more general scope using modular symbols.

The goal of this article is to relate an exponential sum to a version of generalization of Dedekind sums appearing in a paper of Apostol [1] using the cocycle condition. In this setting, the Kloosterman sum appears related to the classical Dedekind sums.

For $i, j \geqslant 1$, we consider the following generalization of Dedekind sums:

$$s_{ij}(p,q) := \sum_{k=0}^{q-1} \bar{B}_i\left(\frac{k}{q}\right)\bar{B}_j\left(\frac{pk}{q}\right),$$

where $\bar{B}_i(x)$ denotes the periodic Bernoulli function. Classical Dedekind sums occur for the case $i = j = 1$. These appeared first in [1] and Carlitz wrote some papers on their properties [7,8]. These vanish for $i + j$ odd after similar reasoning for the vanishing of Bernoulli numbers of odd degree (Corollary 4.2).

There are many ways of writing the Dedekind sums. In this paper, so as to treat the generalized ones as well as the classical, we recover the Dedekind sums as the coefficients of (the germ of) a certain analytic function two variables at 0 in $\mathbb{C}^2$ associated to a 2-dimensional lattice cone in $\mathbb{R}^2$. Similar construction was made by Solomon [36] on a different basis. Garoufalidis and Pommersheim in [13] took the same generating function as ours in defining the generalized Dedekind sums but differ by some power of q. This point will be clarified later in Section 4.

A lattice cone can be identified with an affine linear map $\sigma : [0,1] \to \mathbb{R}^2$ such that $\sigma(0), \sigma(1)$ are two linearly independent primitive lattice vectors. Seen as a singular chain of $\mathbb{R}^2 - 0$, we have obvious notion of the boundary operation. By 1-cocycle, we mean a functional S defined on 2-dimensional cones, which vanishes on the boundary of a (degenerate) 3-dimensional cone. This is equivalent to say that

$$S(\sigma_1) + S(\sigma_2) = S(\sigma)$$

for $\sigma_1 + \sigma_2 = \sigma$ where the addition of cones is defined as their concatenation.

For a pair of relatively prime positive integers p, q (suppose $p < q$ for convenience), one associates a two variable power series denoted by $\mathrm{Todd}_{pq}(x,y)$ defined by Brion–Vergne [6] to write the Euler–Maclaurin summation formula for higher dimensional lattice polytopes (for precise definition, we refer the reader to Section 3.2). $\mathrm{Todd}_{pq}(x,y)$

is the Todd series of the cone $\sigma((1,0),(p,q))$, whose coefficient $\frac{t_{ij}(p,q)}{i!j!}$ of x^iy^j is closely related to $s_{ij}(p,q)$.

In particular, for nonsingular cones, it is

$$\mathrm{Todd}(x,y) = \frac{x}{1-e^{-x}}\frac{y}{1-e^{-y}} = \sum_{i,j}\frac{B_iB_j}{i!j!}x^iy^j. \tag{3}$$

Todd series, after certain normalization, makes a 1-cocycle as functional over chains of 2-dimensional lattice cones, which we call *Todd cocycle* in this article. Using the cocycle condition, we obtain an explicit formula of Todd series of a cone w.r.t. the cone decomposition attached to the continued fraction. Since the cones appearing in continued fraction are nonsingular, the Todd series is decomposed into the Todd series of each nonsingular cones. As seen in (3), we finally obtain an expression of generalized Dedekind sums involving only finite number of Bernoulli numbers. Then multiplied by the denominator which appears to be $R(i,j)q^{N-2}$ which would be 12 for $i=j=1$, after the following theorem we associate certain exponential sums, which generalize the Kloosterman sums.

Theorem 1.1 *(Main theorem). For $i+j=N\geqslant 2$ even, we have*

$$R(i,j)q^{N-2}s_{ij}(p,q) - \frac{p'^{\,i}\alpha_N r_N\binom{N-1}{i} + p^j\alpha_N r_N\binom{N-1}{j}}{q}$$

is always integer where p' is an integer such that $p'p\equiv 1 \pmod q$ and

$$R(i,j) := \binom{N}{i}\beta_N r_N,$$

for an integer r_N (for precise description, see Theorem 5.1) and β_N being the denominator of B_N the N-th Bernoulli number.

Theorem 1.1 relates a particular case of the generalized Kloosterman sum $K_{ij}(k,\ell,q)$ for $k=\alpha_N r_N\binom{N-1}{i}$ and $\ell=\alpha_N r_N\binom{N-1}{j}$ defined below:

Definition 1.2 *(Generalized Kloosterman sum).* For a positive integer q,

$$K_{ij}(k,\ell,q) := \sum_{\substack{0<p<q\\ pp'\equiv 1 \pmod q}} \mathbf{e}\left(\frac{k(p')^i+\ell p^j}{q}\right),$$

where $\mathbf{e}(x) := \exp(2\pi i x)$.

These generalized Kloosterman sums have Weil type bound. It is a consequence of a result on the weight and dimension of the cohomology of an ℓ-adic sheaf due to

Denef–Loeser in [10]. Using the Weil bound for the generalized Kloosterman sums, one can show that the Weyl equidistribution criterion for the generalized Dedekind sums holds.

Thus after our main theorem, the equidistribution property of the (fractional part) of the generalized Dedekind sums multiplied by $R(i,j)q^N$ is similar to that on classical Dedekind sums in [39].

Theorem 1.3. *For even $N = i + j$, the set*

$$\left\{ \left\langle R(i,j)q^{N-2}s_{i,j}(p,q) \right\rangle \mid 0 < p < q,\ (p,q) = 1 \right\}$$

is equidistributed in the interval $[0,1)$, where $\langle x \rangle$ is the fractional part of x in $[0,1)$.

Roughly speaking, the equidistribution of the classical Dedekind sums multiplied by 12 in [39] is a consequence of the modularity of $\eta(\tau)$. Contrary to the classical case, as we don't have such a function which plays the role of $\log \eta(\tau)$ for generalized Dedekind sums, the theorem is not entirely clear from the definition. Thus we argue that this is a consequence of the cocycle property of the Todd series similar to many other properties of the classical Dedekind sums. As mentioned already, for classical Dedekind sums we have $12 = R(1,1)q^0$.

This paper is composed as follows: In Section 2, we review a part of Vardi's result to relate the Kloosterman sum to the classical Dedekind sums. In Section 3, we define the Todd series of a lattice cone and describe the cocycle condition. In Section 4, the generalized Dedekind sums are identified with the coefficients of the Todd series. In Section 5 the generalized Kloosterman sums appear in relation to the generalized Dedekind sums and the proof of Theorem 1.1 is given. Section 6 is devoted to the Weil bound for generalized Kloosterman sums and we finish the proof of the main theorem.

Notations and convention.

- For a real number x, $\langle x \rangle = x - [x]$ is the fractional part taken in $[0,1)$.
- For a function f in x, $f \ll x^{a+\epsilon}$ $(\forall \epsilon > 0)$ means that $|f| = o(x^{a+\epsilon})$ for every positive ϵ.
- $\mathbf{e}(x)$ denotes $\exp(2\pi i x)$.
- The k-th *Bernoulli number* B_i is defined by the generating function

$$\frac{z}{e^z - 1} = \sum_{k=0}^{\infty} \frac{B_k}{k!} z^k.$$

- The k-th *Bernoulli polynomial* $B_k(x)$ is the degree k polynomial defined by

$$\frac{ze^{xz}}{e^z - 1} = \sum_{k=0}^{\infty} \frac{B_k(x)}{k!} z^k.$$

The *k-th periodic Bernoulli function* $\bar{B}_k(x)$ is a $\mathbb{Z}$-periodic function on $\mathbb{R}$ defined by assigning the values for $x \in [0,1)$ as follows:

$$\bar{B}_k(x) = \begin{cases} B_k(\langle x \rangle) & \text{for } x \notin \mathbb{Z}, \\ B_k & \text{for } k \geqslant 2 \text{ and } x \in \mathbb{Z}, \\ 0 & \text{for } k = 1,\ x \in \mathbb{Z}. \end{cases}$$

2. Equidistribution of classical Dedekind sums

In this section, we sketch the proof of the equidistribution of the fractional parts of classical Dedekind sums multiplied by 12 using the Weil bound for Kloosterman sums. The proof appears at the beginning of Vardi [39]. This is not only a special case of the main result of [39] but also a case not covered by the main technic of multiplier system attached to $\log \eta(\tau)$.

As discussed, this step comes from the modularity of $\eta(\tau)$ in relation to $\Delta(\tau)$.

We remind a criterion for a sequence to be equidistributed in $[0,1)$ due to H. Weyl and will show that this is the case. Later, we will be using the Weyl equidistribution criterion for the generalized Dedekind sums.

2.1. Rademacher's theorem

We begin with a reinterpretation of Rademacher's ϕ-function.

Theorem 2.1 *(Rademacher). For a relatively prime pair of integers* (p,q), $12s(p,q) - \frac{p'+p}{q}$ *is always integer whenever* $p'p \equiv 1 \pmod{q}$.

The above Rademacher's theorem is nothing but rephrasing the fact that the values of Rademacher's ϕ-function are taken in $\frac{1}{12}\mathbb{Z}$. At a glance, this is not along the line we follow to show the result of same type for generalized Dedekind sums. But later we will see that the integrality will turn out to be a special case of the cocycle property. Later in Section 5, we will see that the appearance of 12 in the Rademacher theorem is due to the denominator of $B_1 B_2$.

2.2. Kloosterman sums

Let q be a positive integer and k, ℓ be a pair of integers relatively prime to q. The *Kloosterman sum* for k, ℓ of modulus q is denoted by $K(k,\ell,q)$ and defined as

$$K(k,\ell,q) := \sum_{x \in (\mathbb{Z}/q\mathbb{Z})^*} \mathbf{e}\left(\frac{k}{q}x + \frac{\ell}{q}x^{-1}\right) = \sum_{\substack{0 \leqslant x \leqslant q-1 \\ (x,q)=1}} \mathbf{e}\left(\frac{k}{q}x + \frac{\ell}{q}x^{-1}\right).$$

There is a well-known upper bound of Kloosterman sums due to Weil:

$$K(k,\ell,q) \ll q^{\frac{1}{2}+\epsilon} \quad (\forall \epsilon > 0). \tag{4}$$

2.3. Kloosterman sum and Dedekind sum

For two positive integers m and q fixed, if we sum $\mathbf{e}(12ms(p,q))$ over $1 \leqslant p \leqslant q$ such that $(p,q)=1$, we obtain the Kloosterman sum from Theorem 2.1:

$$K(m,m,q) = \sum_{\substack{0<p<q \\ (p,q)=1}} \mathbf{e}\big(12ms(p,q)\big) = \sum_{\substack{0<p<q \\ p'p\equiv 1 \pmod{q}}} \mathbf{e}\left(\frac{p'm+pm}{q}\right). \tag{5}$$

Using Weil's bound (4), we conclude that

$$\sum_{0<q<x} \sum_{\substack{0<p<q \\ (p,q)=1}} \mathbf{e}\big(12ms(p,q)\big) \ll x^{\frac{3}{2}+\epsilon} \quad (\forall \epsilon > 0). \tag{6}$$

Then the above bound implies that the set

$$\big\{\langle 12s(p,q)\rangle \;\big|\; 0<p<q,\ (q,p)=1\big\}$$

fulfills a famous criterion for a sequence in $[0,1)$ to be equidistributed due to H. Weyl [41] as stated below.

2.4. Weyl's equidistribution criterion

A sequence $\{s_i \in [0,1)\}_{i\in\mathbb{N}}$ is equidistributed iff for every $k \in \mathbb{Z}\backslash\{0\}$,

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \mathbf{e}(ks_n) = 0.$$

For a positive integer m and x, let $E(m,x)$ be the value

$$E(m,x) = \frac{1}{\#\{(p,q) \mid \gcd(p,q)=1,\ p<q\leqslant x\}} \sum_{0<q<x} \sum_{\substack{0<p<q \\ (p,q)=1}} \mathbf{e}\big(12ms(p,q)\big).$$

Taking the limit $x \to \infty$, from the Weil bound, we have $E(m,x) \to 0$. Therefore $\langle 12s(p,q)\rangle$ is equidistributed in $[0,1)$.

For the rest of the paper, we consider the possibility of the same sort of equidistribution of the generalized Dedekind sums $s_{ij}(p,q)$.

3. Todd series of a lattice cone

Now we recall the definition of Todd series as defined by Brion–Vergne in [6], for the case of 2-dimensional cones. For 1-dimensional case, this equals the generating function of the Bernoulli numbers up to sign of the variable. The Todd series yields a differential operator of infinite order used in the formulation of the Euler–Maclaurin formula for higher dimensional polytopes. These generating functions are closely related to Shintani functions considered by Solomon in [36] and have similar cocycle property. Later we will see that the Todd series recovers the generalized Dedekind sums as Solomon's Shintani functions. But we warn the reader that these are not the same, *a priori* their cocycle conditions are on the dual cones to each other. This is briefly mentioned by Garoufalidis–Pommersheim in [13]. We will clarify this relation in another paper in sequel [19]. Maybe we could conclude the same result in the framework of Shintani functions. But through our works in relation [18,19], we find it more comfortable to manipulate Todd series rather than Solomon's Shintani functions.

3.1. Lattice cones

Let M be the standard lattice $\mathbb{Z}^2$ in $\mathbb{R}^2$. We consider cones defined in M. By lattice cone, we mean the convex hull of two linearly independent rays of rational slopes. It is always possible to choose unique primitive lattice vectors generating the rays. Let $\sigma = \sigma(v_1, v_2)$ be a lattice cone and v_1, v_2 be primitive lattice generators of the rays bounding σ. Be aware that we take the orientation (i.e. the order of the rays) into consideration so that $\sigma(v_1, v_2) \neq \sigma(v_2, v_1)$. σ is sometimes identified with an integer coefficient matrix A_σ whose columns are the lattice vectors v_1, v_2 in $\mathbb{Z}^2$. M_σ denotes the sublattice of M generated by v_1, v_2. $\Gamma_\sigma = M/M_\sigma$ is isomorphic to a cyclic group of order $|\det(A_\sigma)|$.

For $g \in M$ representing $\gamma \in \Gamma_\sigma$, we have rational numbers $a_{\sigma,i}(g)$, $i = 1, 2$, such that

$$g = a_{\sigma,1}(g)v_1 + a_{\sigma,2}(g)v_2.$$

$a_{i,\sigma}$, being integral on M_σ, yields a character $\chi_{\sigma,i}$ on Γ_σ as

$$\chi_{\sigma,i}(\gamma) := \mathbf{e}\big(a_{\sigma,i}(g)\big), \quad \text{for } i = 1, 2.$$

3.2. Todd series

The Todd power series of σ is defined as:

$$\mathrm{Todd}_\sigma(x_1, x_2) := \sum_{\gamma \in \Gamma_\sigma} \frac{x_1}{1 - \chi_{\sigma,1}(\gamma)e^{-x_1}} \frac{x_2}{1 - \chi_{\sigma,2}(\gamma)e^{-x_2}}. \tag{7}$$

The coefficients of $\mathrm{Todd}_\sigma(x_1, x_2)$ are rational though the expression (7) contains some roots of 1. This is easy to see from the Galois invariance of the expression.

The Todd series is invariant of the $\mathrm{SL}_2(\mathbb{Z})$ equivalent class of cones by the following proposition.

Proposition 3.1. *Let $\sigma = \sigma(v_1, v_2)$ be a lattice cone and A be a matrix in $\mathrm{SL}_2(\mathbb{Z})$. Then we have*

$$\mathrm{Todd}_\sigma(x_1, x_2) = \mathrm{Todd}_{A\sigma}(x_1, x_2).$$

Proof. A gives an isomorphism

$$A : M/M_\sigma \to M/M_{A\sigma}, \quad \gamma \mapsto A\gamma, \quad \text{for } \gamma \in \Gamma_\sigma.$$

For $\gamma = a_{\sigma,1}(\gamma)v_1 + a_{\sigma,2}(\gamma)v_2$,

$$A\gamma = a_{\sigma,1}(\gamma)Av_1 + a_{\sigma,2}(\gamma)Av_2.$$

Therefore

$$\begin{aligned}
\mathrm{Todd}_{A\sigma}(x_1, x_2) &= \sum_{A\gamma \in \Gamma_{A\sigma}} \frac{x_1}{1 - \chi_{A\sigma,1}(A\gamma)e^{-x_1}} \frac{x_2}{1 - \chi_{A\sigma,2}(A\gamma)e^{-x_2}} \\
&= \sum_{A\gamma \in \Gamma_{A\sigma}} \frac{x_1}{1 - \chi_{\sigma,1}(\gamma)e^{-x_1}} \frac{x_2}{1 - \chi_{\sigma,2}(\gamma)e^{-x_2}} \\
&= \mathrm{Todd}_\sigma(x_1, x_2). \qquad \square
\end{aligned}$$

Let $p, q > 0$ be two relatively prime nonnegative integers. Then $(1, 0)$ and (p, q) are primitive lattice vectors and linearly independent. Let σ_{pq} denote the cone generated by $(1, 0)$ and (p, q). Notice that any lattice cone is equivalent to σ_{pq} after basis change. We shall write Todd_{pq} instead of $\mathrm{Todd}_{\sigma_{pq}}$ for simplicity.

3.3. Normalized Todd series and cocycle property

From now on, we will be dealing with only lattice cones in the 1st quadrant. This is not necessary in defining the cocycle property for cones but otherwise we need to extend the category of cones due to the unnecessary occurrence of the Maslov index (cf. [2,19, 36]). For example, in [36], this ambiguity appeared by the name '*formal Cauchy theorem*'. Consequently, for full generality, one has to take the value of the cocycle modulo $\mathbb{Z}$ or to take a central extension of $\mathrm{SL}_2(\mathbb{Z})$ (cf. [4,21]). Here considering only the cones in the 1st quadrant, we can avoid this difficulty. A drawback is that we don't have the cocycles defined over $\mathrm{GL}_2(\mathbb{Q})$ but over singular chains in $\mathbb{R}^2 - 0$. Nevertheless, this won't harm any result of this article.

Definition 3.2. Let σ be a lattice cone. Then the normalized Todd series $S_\sigma(x_1, x_2)$ of σ is defined as

$$S_\sigma(x_1, x_2) = \frac{1}{\det(A_\sigma)x_1x_2}\,\mathrm{Todd}_\sigma(x_1, x_2).$$

Similarly, $S_{\sigma_{pq}}$ is abbreviated to S_{pq} as in unnormalized case.

Because $\mathrm{Todd}_\sigma(x_1, x_2)$ is holomorphic at $0 \in \mathbb{C}^2$, we may well take $\mathrm{Todd}_\sigma(x_1, x_2) \in \mathbb{C}\{\{x_1, x_2\}\}$, where $\mathbb{C}\{\{x_1, x_2\}\}$ is the ring of power series convergent for some neighborhood of 0. Note that the coefficients of $\mathrm{Todd}_\sigma(x_1, x_2)$ lie in $\mathbb{Q}$. Thus we can write

$$\mathrm{Todd}_\sigma(x_1, x_2) \in \mathbb{Q}\{\{x_1, x_2\}\} := \mathbb{C}\{\{x_1, x_2\}\} \cap \mathbb{Q}[[x_1, x_2]].$$

With the notations above, since $S_\sigma(x_1, x_2)$ has simple pole along the two axes: $x_1 = 0$ and $x_2 = 0$,

$$S_\sigma(x_1, x_2) \in \frac{1}{x_1x_2}\mathbb{Q}\{\{x_1, x_2\}\}.$$

Note that swapping two rays of the cone interchanges not only the variables but also the sign in S_σ. Thus the orientation of a cone is reflected in S_σ. In this case, the same cone with the opposite orientation will be denoted by $-\sigma$.

Lemma 3.3. *For a lattice cone σ in the 1st quadrant,*

$$S_{-\sigma}(x_1, x_2) = -S_\sigma(x_2, x_1),$$
$$\mathrm{Todd}_{-\sigma}(x_1, x_2) = \mathrm{Todd}_\sigma(x_2, x_1).$$

Let v_1, v_2, v_3 be pairwise linearly independent primitive lattice vectors in the 1st quadrant. Let σ_{ij} be the lattice cone generated by v_i, v_j. Then we write $\sigma_{ik} = \sigma_{ij} + \sigma_{jk}$. Actually a cone σ generated by lattice vectors v_1, v_2 can be seen as a simplex

$$\sigma : [0, 1] \to \mathbb{R}^2$$

for an affine linear map σ with $\sigma(0) = v_1, \sigma(1) = v_2$. Abusing the notation, if $v_1 = v_2$, such a degenerate cone behaves as a unit when added to $\sigma(v_1, v)$ for any v. Thus we consider the groupoid of 2-dimensional cones. The addition is defined up to boundary of (degenerate) 3-dimensional cones.

Definition 3.4. A 1-*cocycle* over cones is a functional over cones valued in an abelian group M

$$\phi : \sigma \mapsto \phi(\sigma) \in M$$

satisfying $\phi(\sigma_1 + \sigma_2) = \phi(\sigma_1) + \phi(\sigma_2)$.

In terms of groupoids, the 1-cocycle is a map preserving the operation of the groupoid of cones. Our notion of 1-cocycles agrees with the modular pseudo-measures defined by Manin and Marcolli [25], if it were defined on $\mathbb{P}^1_+(\mathbb{R}) := \mathbb{R}^2 - 0/\mathbb{R}^*_+$. Furthermore, the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathbb{R}^2$ makes the 1-cocycle a modular pseudo-measure.

To avoid unnecessary complication, we will consider only the cones lying in the right half plane $\mathbb{R}^2_{x_1\geqslant 0}$. From now on, we denote by '2-d Cones' the set of all 2-dimensional lattice cones in the right half plane. Todd cocycle is a cocycle on 2-d Cones defined as below. Let L be the reduced equation of the line orthogonal to a lattice vector (p,q) for $p>0$. Written explicitly,

$$L = qx_1 - px_2.$$

By $\frac{1}{\prod_L L}\mathbb{Q}\{\{x_1,x_2\}\}$, we mean

$$\sum_L \frac{1}{L}\mathbb{Q}\{\{x_1,x_2\}\}$$

where L runs for all primitive lattice vectors in the right half plane.

Definition–Proposition 3.5 *(Todd cocycle).* The Todd cocycle is a map

$$\Phi : \text{2-d Cones} \to \frac{1}{\prod_L L}\mathbb{Q}\{\{x_1,x_2\}\}$$

given by

$$\Phi(\sigma) = S_\sigma\big(A_\sigma^{-1}(x_1,x_2)\big) \in \mathbb{Q}\{\{x_1,x_2\}\}\big(L_1^{-1},L_2^{-1}\big),$$

where L_i are the equation of lines orthogonal to v_1, v_2. This is a 1-cocycle over 2-dimensional cones in $\mathbb{R}^2$.

Proof. For the proof, we refer the reader to Theorem 3 of [29]. □

4. Generalized Dedekind sum as coefficients of Todd series of a cone

Now we are going to identify the generalized Dedekind sum $s_{ij}(p,q)$ with the $x_1^i x_2^j$-coefficient of Todd_{pq}. We begin with the definition of $s_{ij}(p,q)$. There are variations of the same sum in essential for instance [13,36]. Our convention follows that appeared in p. 71 of [30].

Recall that we consider for positive integers i,j, the generalized Dedekind sum as follows:

$$s_{ij}(p,q) := \sum_{k=0}^{q-1} \bar{B}_i\left(\frac{k}{q}\right)\bar{B}_j\left(\frac{pk}{q}\right).$$

For $i = j = 1$, we have the classical Dedekind sum: $s_{11}(p,q) = s(p,q)$.
Let $\frac{t_{ij}(p,q)}{i!j!}$ be the coefficient of $x_1^i x_2^j$ in $\mathrm{Todd}_{pq}(x_1,x_2)$. Thus

$$\mathrm{Todd}_{pq}(x_1,x_2) = \sum_{i,j\geqslant 0} \frac{t_{ij}(p,q)}{i!j!} x_1^i x_2^j.$$

Theorem 4.1. *(Compare with [13].) We have*

$$t_{ij}(p,q) = -(-q)^{i+j-1}\big(s_{ij}(p,q) + \delta(i,j)B_iB_j\big),$$

where $\delta(i,j) = \begin{cases} 1, & i = 1 \text{ or } j = 1, \\ 0, & \text{otherwise}, \end{cases}$ *and* B_i *is the* i*-th Bernoulli number.*

Proof. Let $\sigma = \sigma_{pq}$ and $v_1 = (1,0)$, $v_2 = (p,q)$ be the primitive nonzero lattice generators of σ. Let v_j^* be the dual vector of v_i for $i = 1,2$ (i.e. $\langle v_i, v_j^*\rangle = \delta_{ij}$). Thus

$$v_1^* = \left(1, -\frac{p}{q}\right) \quad \text{and} \quad v_2^* = \left(0, \frac{1}{q}\right).$$

The dual cone $\check{\sigma}$ of σ is generated by v_1^* and v_2^*.
Then we have

$$\begin{aligned} \mathrm{Todd}_{pq}(x_1,x_2) &= \sum_{g\in\Gamma_{\sigma_{pq}}} \frac{x_1}{1-e^{2\pi i\langle v_1^*,g\rangle}e^{-x_1}} \frac{x_2}{1-e^{2\pi i\langle v_2^*,g\rangle}e^{-x_2}} \\ &= x_1x_2 \sum_{n_1,n_2\geqslant 0} \sum_{g\in\Gamma_{\sigma_{pq}}} e^{2\pi i\langle n_1v_1^*+n_2v_2^*,g\rangle} e^{-n_1x_1-n_2x_2}. \end{aligned} \tag{8}$$

Since

$$\sum_{g\in\Gamma_{\sigma_{p,q}}} e^{2\pi i\langle n_1v_1^*+n_2v_2^*,g\rangle} = \begin{cases} |\Gamma_{\sigma_{p,q}}| = q, & n_1v_1^* + n_2v_2^* \in M^*, \\ 0, & \text{otherwise}, \end{cases}$$

one can write Todd_{pq} as summation over lattice points in $\check{\sigma}$:

$$\begin{aligned} \mathrm{Todd}_{pq}(x_1,x_2) &= qx_1x_2 \sum_{\substack{n_1v_1^*+n_2v_2^*\in M^* \\ n_1,n_2\geqslant 0}} e^{-n_1x_1-n_2x_2} \\ &= qx_1x_2 \sum_{m\in\mathbb{Z}^2\cap\check{\sigma}} e^{-\langle m,v_1\rangle x_1 - \langle m,v_2\rangle x_2}. \end{aligned} \tag{9}$$

Notice that in general v_i^* are not lattice vectors but the primitive lattice generators of $\check{\sigma}$ are

$$u_1 = (q,-p), \qquad u_2 = (0,1).$$

Let $P(u_1, u_2)$ be the following half open parallelogram:

$$P(u_1, u_2) = \{x_1u_1 + x_2u_2 \mid 0 \leqslant x_i < 1\}.$$

Then

$$\check{\sigma} \cap M^* = \big\{z + n_1u_1 + n_2u_2 \;\big|\; z \in P(u_1, u_2) \cap M^*,\ n_i \geqslant 0\big\}.$$

Thus we have

$$\begin{aligned}\mathrm{Todd}_{pq}(x_1, x_2) &= qx_1x_2 \sum_{z\in P(u_1,u_2)\cap M^*} \frac{e^{-\langle z,v_1\rangle x_1 - \langle z,v_2\rangle x_2}}{(1-e^{-qx_1})(1-e^{-qx_2})} \\ &= q^{-1} \sum_{i,j\geqslant 0} \sum_{z\in P(u_1,u_2)\cap M^*} B_i\left(\frac{\langle z, v_1\rangle}{q}\right) B_j\left(\frac{\langle z, v_2\rangle}{q}\right) (-q)^{i+j} \frac{x_1^i x_2^j}{i!j!}. \qquad (10)\end{aligned}$$

The lattice points inside $P(u_1, u_2)$ are identified as follows:

$$P(u_1, u_2) \cap M^* = \left\{\frac{k}{q}u_1 + \left\langle \frac{pk}{q}\right\rangle u_2 \;\middle|\; k = 0, 1, 2, \ldots, q-1\right\}.$$

Hence $t_{ij}(p, q)$ and $s_{ij}(p, q)$ are related in the desired form:

$$\begin{aligned}t_{ij}(p,q) &= q^{-1}(-q)^{i+j} \sum_{z\in P(u_1,u_2)\cap M^*} B_i\left(\frac{\langle z, v_1\rangle}{q}\right) B_j\left(\frac{\langle z, v_2\rangle}{q}\right) \\ &= q^{-1}(-q)^{i+j} \sum_{k=0}^{q-1} B_i\left(\frac{k}{q}\right) B_j\left(\left\langle \frac{pk}{q}\right\rangle\right) \\ &= q^{-1}(-q)^{i+j} \left(\sum_{k=0}^{q-1} \overline{B}_i\left(\frac{k}{q}\right) \overline{B}_j\left(\frac{pk}{q}\right) + \delta(i,j) B_iB_j\right) \\ &= -(-q)^{i+j-1}\big(s_{ij}(p,q) + \delta(i,j)B_iB_j\big). \qquad \square\end{aligned}$$

For odd $i + j$, $s_{ij}(p, q)$ will turn out to be trivial. This is easy consequence of the previous theorem.

Let us define $L^\lambda(x)$ for a fixed complex number $\lambda \neq 0$ as

$$L^\lambda(x) := \frac{x}{2}\frac{1+\lambda e^{-x}}{1-\lambda e^{-x}}.$$

For $\lambda = 1$, this is the even part of $\mathrm{Todd}(x)$:

$$\mathrm{Todd}(x) := \frac{x}{1-e^{-x}} = \frac{x}{2} + L^{\lambda=1}(x).$$

For $\lambda \neq 1$, $L^{\lambda}(x)$ is not even in general, but we have

$$\mathrm{Todd}^{\lambda}(x) := \frac{x}{1-\lambda e^{-x}} = \frac{x}{2} + L^{\lambda}(x) \tag{11}$$

and

$$L^{\lambda}(-x) = L^{\lambda^{-1}}(x). \tag{12}$$

Thus if σ is a lattice cone, $\sum_{g\in\Gamma_\sigma} L^{\chi_i(g)}(x_i)$ is even for $i = 1, 2$. Therefore we have a decomposition of $\mathrm{Todd}_\sigma(x_1, x_2)$ as follows:

$$\begin{aligned}\mathrm{Todd}_\sigma(x_1,x_2) = \frac{q}{4}x_1x_2 + \frac{1}{2}\sum_{g\in\Gamma_\sigma}\big(x_1L^{\chi_2(g)}(x_2) + x_2L^{\chi_1(g)}(x_1)\big)\\ + \frac{1}{4}\sum_{g\in\Gamma_\sigma} L^{\chi_1(g)}(x_1)L^{\chi_2(g)}(x_2).\end{aligned} \tag{13}$$

Notice that the odd part of $\mathrm{Todd}_\sigma(x_1, x_2)$ is

$$\frac{1}{2}\sum_{g\in\Gamma_\sigma}\big(x_1L^{\chi_2(g)}(x_2) + x_2L^{\chi_1(g)}(x_1)\big).$$

So $t_{ij}(p, q) = 0$ for $i + j$ odd and $i, j > 1$. Otherwise, for example $i = 1$ and $j = 2k$,

$$t_{1,2k}(p,q) = \frac{q^{2k}}{2}B_{2k} = -q^{2k}\big(s_{1,2k}(p,q) + B_1B_{2k}\big).$$

Since $B_1 = -\frac{1}{2}$, again we have $s_{1,2k}(p, q) = 0$.

Hence we obtain the following corollary:

Corollary 4.2. *Let p, q be relatively prime pair of integers. Then for given $i, j \geqslant 1$ such that $i + j$ is odd,*

$$s_{ij}(p,q) = 0.$$

For the rest of this paper, we assume $i + j$ is even.

5. Generalized Dedekind sums and generalized Kloosterman sums

In this section, we would like to evaluate $s_{ij}(p, q)$ in terms of the continued fraction of q/p for even $i + j = N$. As we saw in the previous section, $s_{ij}(p, q)$ vanishes if $i + j$ is odd.

Writing explicitly $s_{ij}(p, q)$, we will obtain an analogous statement to Rademacher's theorem (Theorem 2.1). Then we will be able to relate the generalized Kloosterman sums to a generalization of Dedekind sums.

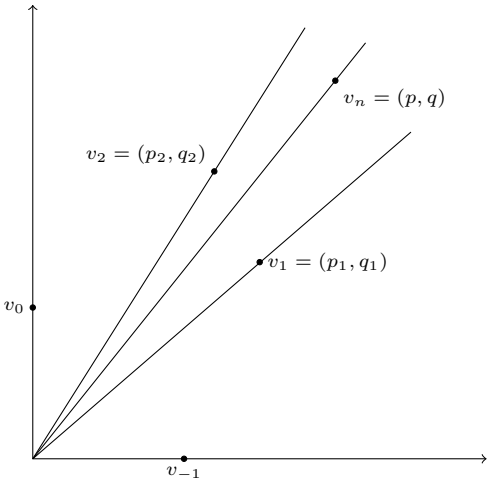


Fig. 1. ν_i.

The geometric counterpart of the continued fraction is the cone decomposition (cf. [17,38]). Accordingly, the (normalized) Todd series is decomposed into sum of the (normalized) Todd series of nonsingular cones.

Let q and p be relatively prime positive integers and suppose $q > p$.

We are going to associate the (positive) continued fraction of q/p:

$$\frac{q}{p} = a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}},$$

where $a_i \geqslant 1$ are all integers. We put $(p_{-1}, q_{-1}) = (1, 0)$, $(p_0, q_0) = (0, 1)$ and for $i \geqslant 1$. Define a pair of relatively prime integers p_i and q_i using truncation of the continued fraction of $\frac{q}{p}$:

$$\frac{q_i}{p_i} := a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_i}}}.$$

As previously, let $\sigma := \sigma_{p,q}$ and v_k be the primitive lattice vectors in the 1st quadrant (p_k, q_k) for $-1 \leqslant k \leqslant n$ (see Fig. 1).

Then we have the following virtual cone decomposition of σ into nonsingular cones:

$$\sigma := \sigma_{pq} = \sigma(v_{-1}, v_n) = \sum_{k=-1}^{n-1} \sigma_k,$$

where $\sigma_k := \sigma(v_k, v_{k+1})$.

After the additivity of normalized Todd series, according to the continued fraction of q/p, we obtain the following expression:

$$S_{pq}(x,y) = \sum_{k=-1}^{n-1} (-1)^{k+1} F\big(A_{\sigma_k}^{-1} A_\sigma (x,y)^t\big), \tag{14}$$

where

$$F(x,y) = \frac{1}{1-e^{-x}} \frac{1}{1-e^{-y}} = \frac{\mathrm{Todd}(x,y)}{xy}.$$

One should note that $F(x,y)$ is the normalized Todd series of a nonsingular cone (up to sign).

Recall that the 1-variable Todd series is

$$\mathrm{Todd}(z) = \frac{z}{1-e^{-z}} = \sum_{i=0}^{\infty} (-1)^i \frac{B_i}{i!} z^i.$$

The matrix $A_{\sigma_k}^{-1} A_\sigma$ is computed as

$$A_{\sigma_k}^{-1} A_\sigma = (-1)^{k+1} \begin{pmatrix} q_{k+1} & pq_{k+1} - qp_{k+1} \\ -q_k & -pq_k + qp_k \end{pmatrix}.$$

As $\det(A_\sigma) = q$, by multiplying qxy we obtain the following expression of Todd series of σ from (14):

$$\begin{aligned} \mathrm{Todd}_{pq}(x,y) &= qxy \sum_{k=-1}^{n-1} (-1)^{k+1} \frac{\mathrm{Todd}(M_k)\,\mathrm{Todd}(M_{k+1})}{M_k M_{k+1}} \\ &= qxy \sum_{k=-1}^{n-1} (-1)^{k+1} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} (-1)^{i+j} \frac{B_i}{i!} \frac{B_j}{j!} M_k^{j-1} M_{k+1}^{i-1}, \end{aligned} \tag{15}$$

where

$$M_k := \begin{cases} qy, & k = -1, \\ (-1)^k (q_k x + (pq_k - qp_k) y), & 0 \leqslant k \leqslant n-1, \\ (-1)^n qx, & k = n. \end{cases} \tag{16}$$

Denote by Todd_σ^N the degree N homogeneous part of Todd_σ. Then from (15) Todd_σ^N is given as follows:

$$\begin{aligned} \mathrm{Todd}_\sigma^N &= qxy \sum_{k=-1}^{n-1} (-1)^{k+1} \sum_{i=0}^{N-2} (-1)^N \frac{B_{i+1}}{(i+1)!} \frac{B_{N-i-1}}{(N-i-1)!} M_k^{N-2-i} M_{k+1}^i \\ &\quad + (-1)^N qxy \frac{B_N}{N!} \sum_{k=-1}^{n-1} (-1)^{k+1} \frac{M_k^N + M_{k+1}^N}{M_k M_{k+1}}. \end{aligned} \tag{17}$$

Since for $k \geqslant 0$ we have

$$M_{k-1} - M_{k+1} = a_{k+1} M_k,$$

$$\begin{aligned} &qxy \sum_{k=-1}^{n-1} (-1)^{k+1} \frac{M_k^N + M_{k+1}^N}{M_k M_{k+1}} \\ &\quad = qxy \left(\sum_{k=0}^{n-1} (-1)^k a_{k+1} \sum_{i=0}^{N-2} M_{k-1}^{N-2-i} M_{k+1}^i \right) + M_0^{N-1} x + M_{n-1}^{N-1} y. \end{aligned} \tag{18}$$

Therefore, plugging (18) into (17), we obtain

$$\begin{aligned} \operatorname{Todd}_{pq}^N = {} & qxy \sum_{k=-1}^{n-1} (-1)^{k+1} \sum_{i=0}^{N-2} (-1)^N \frac{B_{i+1}}{(i+1)!} \frac{B_{N-i-1}}{(N-i-1)!} M_k^{N-2-i} M_{k+1}^i \\ & + (-1)^N q \frac{B_N}{N!} xy \left(\sum_{k=0}^{n-1} (-1)^k a_{k+1} \sum_{i=0}^{N-2} M_{k-1}^{N-i-2} M_{k+1}^i \right) \\ & + \frac{B_N}{N!} M_0^{N-1} x + \frac{B_N}{N!} M_{n-1}^{N-1} y. \end{aligned} \tag{19}$$

Consequently, we obtain the following integrality involving $\operatorname{Todd}_{pq}^N$.

Theorem 5.1. *Suppose N is an even positive integer. Let $\frac{\alpha_k}{\beta_k}$ be the reduced fraction of $B_k \neq 0$ with $\beta_k > 0$ and*

$$r_N := \mathit{L.C.M.} \left\{ \mathit{Denominator\ of\ } \beta_N \binom{N}{i+1} B_{i+1} B_{N-i-1} \;\middle|\; i \text{ odd},\ 0 \leqslant i \leqslant N-2 \right\}.$$

Then we have

$$\begin{aligned} & N! \beta_N r_N \operatorname{Todd}_{pq}^N(x,y) \\ & \quad - \alpha_N r_N (x+py)^{N-1} x - \alpha_N r_N \big((-1)^{n-1} q_{n-1} x + y \big)^{N-1} y \in q\mathbb{Z}[x,y]. \end{aligned}$$

Proof. By multiplying $N! \beta_N$ on Eq. (19), we have that

$$\begin{aligned} & N! \beta_N \operatorname{Todd}_{pq}^N - \alpha_N M_0^{N-1} x - \alpha_N M_{n-1}^{N-1} y \\ & \quad = qxy \sum_{k=-1}^{n-1} (-1)^{k+1} \sum_{i=0}^{N-2} (-1)^N \beta_N \binom{N}{i+1} B_{i+1} B_{N-i-1} M_k^{N-i-2} M_{k+1}^i \\ & \qquad + (-1)^N q \alpha_N xy \left(\sum_{k=0}^{n-1} (-1)^k a_{k+1} \sum_{i=0}^{N-2} M_{k-1}^{N-2-i} M_{k+1}^i \right). \end{aligned} \tag{20}$$

Since $M_i \in \mathbb{Z}[x,y]$ for every i, multiplying (20) by r_N, we conclude the proof. □

Proof of Theorem 1.1. If we read coefficient of $x^i y^{N-i}$ in the previous theorem, we obtain the formula of Theorem 1.1. □

Remark 5.2. In Theorem 1.1, if we consider the case of $i = 1 = j$, so $N = 2$, then the denominator β_2 of B_2 is 6 and r_N is 1. Thus, we have $R(1,1) = 12$. This is the case of Theorem 2.1 considered by Rademacher.

According to Theorem 1.1, we associate the generalized Kloosterman sums to the generalized Dedekind sums as follows:

$$\begin{aligned}
&\sum_{\substack{0<p<q\\(p,q)=1}} \mathbf{e}\big(R(i,j)q^{N-2}s_{ij}(p,q)\big)\\
&= \sum_{\substack{0<p<q\\pp'\equiv 1 \pmod q}} \mathbf{e}\left(\frac{(p')^i\alpha_N r_N\binom{N-1}{i} + p^j\alpha_N r_N\binom{N-1}{j}}{q}\right)\\
&= K_{ij}\left(\alpha_N r_N\binom{N-1}{i}, \alpha_N r_N\binom{N-1}{j}, q\right). \qquad (21)
\end{aligned}$$

6. Bounds for generalized Kloosterman sums

In this section, we are going to investigate the Weil type bound for the generalized Kloosterman sums $K_{ij}(k,\ell,q)$. This estimate amounts basically to asking the weight of the cohomology of a certain ℓ-adic sheaf. From the bound we will show that the Weyl equidistribution criterion for $\{\langle R(i,j)q^N s_{ij}(p,q)\rangle\}$ is fulfilled for $i+j$ even. This will conclude the proof of the equidistribution of the generalized Dedekind sums.

6.1. Weight of ℓ-adic sheaf and exponential sums

Let us first recall the work of Denef–Loeser [10]. Let X be a scheme of finite type over $k := \mathbb{F}_q$ and lets $\psi : k \to \mathbb{C}^*$ be a nontrivial additive character. Then a $\overline{\mathbb{Q}}_\ell$-sheaf $\mathscr{L}_\psi$ on $\mathbb{A}^1_k$ is associated to ψ and the Artin–Schreier covering $t^q - t = x$. For a morphism $f : X \to \mathbb{A}^1_k$, the exponential sum

$$S(f) = \sum_{x\in X(k)} \psi\big(f(x)\big)$$

is defined. Let Fr denote the (geometric) Frobenius action. Grothendieck's trace formula identifies this exponential sum with the trace of the Frobenius action on the cohomology:

$$S(f) = \sum_i (-1)^i \operatorname{Tr}\big(Fr^* | H^i_c\big(X\otimes\bar{k}, f^*\mathscr{L}_\psi\big)\big).$$

For $X = T_k^n$, if a map $f : X \to A_k^1$ is given by a Laurent polynomial $f = \sum_{i\in\mathbb{Z}^n} c_i x^i$, the Newton polyhedron $\Delta_\infty(f)$ is defined as the convex hull of $\{i \in \mathbb{Z}^n \mid c_i \neq 0\}$ in $\mathbb{R}^n$. f is said to be *non-degenerate w.r.t.* $\Delta_\infty(f)$ if for every face σ of $\Delta_\infty(f)$ that does not contain 0, the locus

$$\frac{\partial f_\sigma}{\partial x_1} = \cdots = \frac{\partial f_\sigma}{\partial x_n} = 0$$

is empty. Then a result of Denef and Loeser (Theorem 1.3 in [10]) is stated as follows:

Theorem 6.1. *(See Denef–Loeser [10].) Suppose $f : T_k^n \to \mathbb{A}_k^1$ is non-degenerate w.r.t. $\Delta_\infty(f)$ and $\dim \Delta_\infty(f) = n$. Then we have*

(1) $H_c^i(T_{\bar{k}}^n, f^*\mathscr{L}_\psi) = 0$ *for* $i \neq n$,
(2) $\dim H_c^n(T_{\bar{k}}^n, f^*\mathscr{L}_\psi) = n!\,\mathrm{Vol}(\Delta_\infty(f))$.

If moreover the interior of $\Delta_\infty(f)$ contains 0, then

(3) $H_c^n(T_k^n, f^*\mathscr{L}_\psi)$ *is pure of weight n (i.e. all Frobenius eigenvalues have absolute value $q^{n/2}$).*

If f satisfies the conditions of the above theorem, the trace formula is simplified as

$$S(f) = (-1)^n \operatorname{Tr}\big(Fr^* | H_c^n\big(X \otimes \bar{k}, f^*\mathscr{L}_\psi\big)\big).$$

Then the Weil type bound is a simple consequence of the purity result:

$$\big|S(f)\big| \leqslant \sum |\text{Frobenius eigenvalue}| \leqslant C_f q^{n/2},$$

where $C_f = \dim H_c^n(T_{\bar{k}}^n, f^*\mathscr{L}_\psi) = n!\,\mathrm{Vol}(\Delta_\infty(f))$.

By the fundamental result of Deligne in [9], we know that $H_c^n(T_{\bar{k}}^n, f^*\mathscr{L}_\psi)$ has mixed weight $\leqslant n$. This is already enough to obtain the Weil bound, but to obtain the dimension, we need the theorem of Denef–Loeser.

First, for $q = p$ and $f(z) = kz^i + \ell z^{-j}$ non-degenerate, we obtain the Weil bound for the generalized Kloosterman sum. Second, we reduce the generalized Kloosterman sum of composite modulus to a product of those of p-primary modulus. Besides, we will consider those exceptional cases separately. Altogether, this bound yields the Weyl criterion for equidistribution.

6.2. Reduction to non-degenerate case

Let us write first the Weil bound for the generalized Kloosterman sum of prime modulus in non-degenerate case.

Lemma 6.2 *(Non-degenerate case). Let p be a prime and p do not divide i and j. Suppose that k and ℓ are not divisible by p. Then we have*

$$\left|K_{ij}(k,\ell,p)\right| \leqslant (i+j)p^{1/2}.$$

Proof. The condition on i,j,k,ℓ ensures that $f(z)=kz^i+\ell z^{-j}$ non-degenerate w.r.t. its Newton polyhedron. It is a direct consequence of Theorem 6.1 due to Denef–Loeser. □

Suppose that f fails to be non-degenerate for a given prime p. This happens when p divides at least one of i, j, k or ℓ. *A priori* f can be degenerated for only finitely many cases of prime p. Since we vary p for fixed i,j, it is easy to see

$$\left|K_{ij}(k,\ell,p)\right| \leqslant Cp^{1/2}$$

for a constant C independent of p but determined by i and j.

If q has many prime factors, we need to reduce the case to non-degenerate. This will be justified through the next two lemmas.

First we consider the case q being power of a prime p. When either k or ℓ is divisible by some power of p, the first reduction is as follows:

Lemma 6.3. *Let p be a fixed prime and $k=k'p^\beta$, $\ell=\ell'p^\beta$ for $p^\beta \| \gcd(k,\ell)$. Then for given i and j, we have*

$$K_{ij}\big(k,\ell,p^\alpha\big) = p^\beta K_{ij}\big(k',\ell',p^{\alpha-\beta}\big).$$

Proof. We note that an element $z\in(\mathbb{Z}/p^\alpha\mathbb{Z})^*$ is expressed as

$$z = p^{\alpha-\beta}x+y$$

for $x\in\mathbb{Z}/p^\beta\mathbb{Z}$ and $y\in(\mathbb{Z}/p^{\alpha-\beta}\mathbb{Z})^*$.

Thus, we find that

$$\begin{aligned} K_{ij}\big(k,\ell,p^\alpha\big) &= \sum_{z\in(\mathbb{Z}/p^\alpha\mathbb{Z})^*} \mathbf{e}\left(\frac{kz^i+\ell z^{-j}}{p^\alpha}\right) = \sum_{x\in\mathbb{Z}/p^\beta\mathbb{Z}}\sum_{y\in(\mathbb{Z}/p^{\alpha-\beta}\mathbb{Z})^*} \mathbf{e}\left(\frac{k'y^i+\ell' y^{-j}}{p^{\alpha-\beta}}\right) \\ &= p^\beta K_{ij}\big(k',\ell',p^{\alpha-\beta}\big). \qquad \square \end{aligned} \tag{22}$$

After the previous lemma, we can pull out p-factors out of k,ℓ. For non-degenerate k and ℓ, we obtain the following bound:

Lemma 6.4. *Let p be a prime and suppose at least one of k or ℓ is indivisible by p. Then, for given positive integers i,j*

$$\left|K_{ij}\big(k,\ell,p^\alpha\big)\right| \leqslant ij(i+j)^{\frac{3}{2}}p^{\frac{\alpha}{2}}.$$

Proof. Applying Lemma 12.2–3 of [16] for the cases α even and odd, we can reduce the estimation to that of simpler sums respectively.

For $\alpha = 2\beta$, we have

$$\begin{aligned}\left|K_{ij}\left(k,\ell,p^{2\beta}\right)\right| &= p^{\beta}\left|\sum_{\substack{x\in(\mathbb{Z}/p^{\beta}\mathbb{Z})^{*}\\ ikx^{i+j}=j\ell}} \mathbf{e}\left(\frac{kx^{i}+\ell x^{-j}}{p^{2\beta}}\right)\right| \\ &\leqslant p^{\beta}\sum_{\substack{x\in(\mathbb{Z}/p^{\beta}\mathbb{Z})^{*}\\ ikx^{i+j}=j\ell}} 1 \leqslant p^{\beta} ij(i+j). \end{aligned} \tag{23}$$

For $\alpha = 2\beta+1$,

$$K_{ij}\left(k,\ell,p^{2\beta+1}\right) = p^{\beta}\sum_{\substack{x\in(\mathbb{Z}/p^{\beta}\mathbb{Z})^{*}\\ ikx^{i+j}=j\ell}} \mathbf{e}\left(\frac{kx^{i}+\ell x^{-j}}{p^{2\beta+1}}\right) G_p(x) \tag{24}$$

where

$$G_p(x) = \sum_{y\in\mathbb{Z}/p\mathbb{Z}} e_p\left(d(x)y^2 + h(x)p^{-\beta}y\right).$$

Here,

$$\begin{aligned} e_p(x) &= \mathbf{e}\left(\frac{x}{p}\right), \\ d(x) &= \frac{1}{2}\left(ki(i-1)x^{i-2} + \ell j(j+1)x^{-j-2}\right) \quad \text{and} \\ h(x) &= kix^{i-1} - \ell j x^{-j-1}. \end{aligned}$$

Note that

$$G_p(x) = \begin{cases} p & \text{for } 2d(x)\equiv 0,\ h(x)p^{-\beta}\equiv 0 \pmod p, \\ 0 & \text{for } 2d(x)\equiv 0,\ h(x)p^{-\beta}\not\equiv 0 \pmod p, \end{cases}$$

and $|G_p(x)| \leqslant \sqrt{p}$ for $2d(x)\not\equiv 0 \pmod p$. If $2d(x)\equiv h(x)p^{-\beta}\equiv 0 \pmod p$, then p divides $j(i+j)$, thus $p\leqslant i+j$. Thus we obtain a general bound for $G_p(x)$:

$$\left|G_p(x)\right| \leqslant p^{\frac{1}{2}}(i+j)^{\frac{1}{2}}.$$

This yields the desired bound for odd α. □

These two lemmas imply the Weil bound for $q = p^{\alpha}$ as follows:

Proposition 6.5. *For all positive integers i, j and positive prime p, $q = p^\alpha$, we have*

$$\left|K_{ij}\left(k,\ell,p^\alpha\right)\right| \leqslant ij(i+j)^{\frac{3}{2}}\left(k,\ell,p^\alpha\right)^{\frac{1}{2}}p^{\frac{\alpha}{2}}.$$

Now we need reduction to a single prime factor when there are several prime factors of q.

Lemma 6.6. *Let $q_1 > 1$ and $q_2 > 1$ be relatively prime integers. For k, ℓ, let k_i and ℓ_i be the mod q_i Chinese remainder (i.e. under the isomorphism $\mathbb{Z}/q_1q_2\mathbb{Z} \to \mathbb{Z}/q_1 \times \mathbb{Z}/q_2\mathbb{Z}$, $k \mapsto (k_1, k_2)$ and $\ell \mapsto (\ell_1, \ell_2)$). Then we have*

$$K_{ij}(k,\ell,q_1q_2) = K_{ij}(k_1,\ell_1,q_1)K_{ij}(k_2,\ell_2,q_2).$$

Proof. This is an easy consequence of Fubini's theorem. □

Since $\sum_{q<x}\phi(q) \sim x^2$ for the Euler-phi function ϕ, we come to the proof of the main theorem from the Weyl criterion for equidistribution and forthcoming Proposition 6.7.

Combining Proposition 6.5 and Lemma 6.6, we have the following result:

Proposition 6.7. *For all positive integers i, j and q,*

$$\left|K_{ij}(k,\ell,q)\right| \leqslant \left(ij(i+j)^{\frac{3}{2}}\sqrt{(k,\ell)}\right)^{\omega(q)}\sqrt{q},$$

where $\omega(q)$ is the number of prime factors of q.

Proof. Let $q = p_1^{n_1}p_2^{n_2}\cdot\cdots\cdot p_{\omega(q)}^{n_{\omega(q)}}$, for distinct primes $p_1, p_2, \ldots, p_{\omega(q)}$.

Since $(k,\ell,p^\alpha) \leqslant (k,\ell)$, Proposition 6.5 directly implies that for any prime p,

$$\left|K_{i,j}\left(k,\ell,p^\alpha\right)\right| \leqslant ij(i+j)^{\frac{3}{2}}\sqrt{(k,\ell)}\sqrt{p^\alpha}.$$

Finally, after the multiplicativity of Kloosterman sum as in Lemma 6.6, Proposition 6.7 is proved. □

6.3. Proof of Theorem 1.3

Finally, we deduce the Weyl criterion from the bound of the generalized Kloosterman sums.

$\omega(q)$ in Proposition 6.7 has well-known estimate:

$$\omega(q) \sim \log\log q. \tag{25}$$

For sufficiently large q,

$$\left(ij(i+j)\sqrt{(k,\ell)}\right)^{\omega(q)} \leqslant \left(ij(i+j)^{\frac{3}{2}}\sqrt{(k,\ell)}\right)^{c\log\log q} \leqslant (\log q)^{c\log(i+j)^{\frac{3}{2}}ij\sqrt{(k,\ell)}}.$$

Thus, we have that for any $\epsilon > 0$,

$$\left(ij(i+j)^{\frac{3}{2}}\sqrt{(k,\ell)}\right)^{\omega(q)} \ll q^{\epsilon}.$$

Therefore, we have the following Weil type bound:

Theorem 6.8. *For given pair of positive integers i, j,*

$$\left|K_{ij}(k,\ell,q)\right| \ll q^{\frac{1}{2}+\epsilon}, \quad \forall \epsilon > 0.$$

Now, we show the Weyl criterion of generalized Dedekind sum from the following estimate:

$$\begin{aligned}&\sum_{0<q<x}\sum_{\substack{0<p<q\\(p,q)=1}} \mathbf{e}\big(mR(i,j)q^{N-2}s_{i,j}(p,q)\big)\\ &\quad= \sum_{0<q<x} K_{i,j}\left(m\alpha_N r_N\binom{N-1}{i}, m\alpha_N r_N\binom{N-1}{j}, q\right) \leqslant x^{\frac{3}{2}+\epsilon}. \end{aligned} \tag{26}$$

Consequently, Weyl's equidistribution criterion is fulfilled for the fractional part of $R(i,j)q^{N-2}s_{i,j}(p,q)$:

$$\begin{aligned}E_{ij}(m,x) = &\frac{1}{\#\{(p,q) \mid \gcd(p,q)=1,\ p<q\leqslant x\}}\\ &\times \sum_{0<q<x}\sum_{\substack{0<p<q\\(p,q)=1}} \mathbf{e}\big(mR(i,j)q^{N-2}s_{i,j}(p,q)\big) \to 0,\end{aligned} \tag{27}$$

as $x \to \infty$. Therefore the proof of the main theorem is finished.

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References

[1] T.M. Apostol, Generalized Dedekind sums and transformation formulae of certain Lambert series, Duke Math. J. 17 (1950) 147–157.
[2] V.I. Arnold, A characteristic class entering in quantization conditions, Funct. Anal. Appl. 1 (1967) 1–14.
[3] T. Asai, The reciprocity of Dedekind sums and the factor set for the universal covering group of $SL_2(\mathbb{R})$, Nagoya Math. J. 37 (1970) 67–80.

[4] M.F. Atiyah, The logarithm of Dedekind η-function, Math. Ann. 278 (1–4) (1987) 335–380.
[5] P. Barkan, Sur les sommes de Dedekind et les fractions continues finies, C. R. Acad. Sci. Paris Sér. A–B 284 (16) (1977) A923–A926.
[6] M. Brion, M. Vergne, Lattice points in simple polytopes, J. Amer. Math. Soc. 10 (1997) 371–392.
[7] L. Carlitz, Some theorems on generalized Dedekind sums, Pacific J. Math. 3 (1953) 513–522.
[8] L. Carlitz, Linear relations among generalized Dedekind sums, J. Reine Angew. Math. 280 (1965) 154–162.
[9] P. Deligne, La conjecture de Weil II, Publ. Math. Inst. Hautes Études Sci. 52 (1980) 137–252.
[10] J. Denef, F. Loeser, Weights of exponential sums, intersection cohomology and Newton polyhedra, Invent. Math. 106 (2) (1991) 275–294.
[11] S. Fukuhara, Modular forms, generalized Dedekind symbols and period polynomials, Math. Ann. 310 (1998) 83–101.
[12] S. Fukuhara, Dedekind symbols with plus reciprocity laws, J. Number Theory 128 (4) (2008) 781–795.
[13] S. Garoufalidis, J. Pommersheim, Values of zeta functions at negative integers, Dedekind sums and toric geometry, J. Amer. Math. Soc. 14 (1) (2000) 1–23.
[14] D. Hickerson, Continued fractions and density results for Dedekind sums, J. Reine Angew. Math. 290 (1977) 113–116.
[15] F. Hirzebruch, The signature theorem: reminiscences and recreation, in: Prospects in Mathematics, Proc. Sympos., Princeton Univ., Princeton, NJ, 1970, in: Ann. of Math. Stud., vol. 70, Princeton Univ. Press, Princeton, NJ, 1971, pp. 3–31.
[16] H. Iwaniec, E. Kowalski, Analytic Number Theory, Amer. Math. Soc. Colloq. Publ., vol. 53, AMS, 2004.
[17] B. Jun, J. Lee, The behavior of Hecke's L-functions of real quadratic fields at $s = 0$, Algebra Number Theory 5 (8) (2011) 1001–1026.
[18] B. Jun, J. Lee, Special values of partial zeta functions of real quadratic fields at nonpositive integers and Euler–Maclaurin formula, arXiv:1209.4958, 2012.
[19] B. Jun, J. Lee, Reciprocity law of generalized Dedekind sums from Todd cocycle, preprint.
[20] B. Jun, J. Lee, Equidistribution of generalized Dedekind sums and integrality of special values of partial zeta functions at non-positive integers, in preparation.
[21] R. Kirby, P. Melvin, Dedekind sums, μ-invariants and the signature cocycle, Math. Ann. 299 (2) (1994) 231–267.
[22] D. Knuth, Notes on generalized Dedekind sums, Acta Arith. 33 (1973) 297–325.
[23] Yu.I. Manin, Parabolic points and zeta-functions of modular curves, Math. USSR Izv. 6 (1) (1972) 19–64; in: Selected Papers, World Scientific, 1996, pp. 202–247.
[24] Yu.I. Manin, Non-commutative generalized Dedekind symbols, arXiv:1301.0078.
[25] Yu.I. Manin, M. Marcolli, Modular shadows and the Lévy–Mellin ∞-adic transform, in: Modular Forms on Schiermonnikoog, Cambridge Univ. Press, Cambridge, 2008, pp. 189–238.
[26] C. Meyer, Über die Berechnung der Klassen zahl abelscher Körper quadratischen Zahlkörpern, Akademie-Verlag, Berlin, 1955.
[27] W. Meyer, Die Signatur von Flächenbündeln, Math. Ann. 256 (1981) 255–276.
[28] G. Myerson, Dedekind sums and uniform distribution, J. Number Theory 28 (1988) 233–239.
[29] J. Pommersheim, Barvinok's algorithm and the Todd class of a toric variety, J. Pure Appl. Algebra 117, 118 (1997) 519–533.
[30] H. Rademacher, E. Grosswald, Dedekind Sums, Carus Math. Monogr., vol. 16, Math. Assoc. Amer., 1972.
[31] R. Sczech, Dedekind sums and signatures of intersection forms, Math. Ann. 299 (1) (1994) 269–274.
[32] A. Selberg, On the Estimation of Fourier Coefficients of Modular Forms, Proc. Sympos. Pure Math., vol. VIII, Amer. Math. Soc., Providence, RI, 1965.
[33] T. Shintani, On evaluation of zeta functions of totally real algebraic number fields at non-positive integers, J. Fac. Sci. Univ. Tokyo 23 (1976) 393–417.
[34] C.L. Siegel, Lectures on Advanced Analytic Number Theory, Tata Inst., Bombay, 1961.
[35] C.L. Siegel, Berechnung von Zetafunktionen an ganzzahligen Stellen, Nachr. Akad. Wiss. Göttingen Math.-Phys. Kl. 10 (1969) 87–102.
[36] D. Solomon, Algebraic properties of Shintani generating functions: Dedekind sums and cocycles on $\mathrm{PGL}(2,\mathbb{Q})$, Compos. Math. 12 (3) (1998) 333–364.
[37] G. Stevens, The Eisentein measure and real quadratic fields, in: J.M. de Koninck, C. Levesque (Eds.), Proceedings of the International Number Theory Conference, Université Laval, 1987, de Gruyter, 1989.

[38] G. van der Geer, Hilbert Modular Surfaces, Ergeb. Math. Grenzgeb. (3), vol. 16, Springer, 1988.
[39] I. Vardi, A relation between Dedekind sums and Kloosterman sums, Duke Math. J. 55 (1) (1987) 189–197.
[40] A. Weil, On some exponential sums, Proc. Nat. Acad. Sci. 34 (1948) 204–207.
[41] H. Weyl, Über die Gleichverteilung von Zahlen mod. Eins, Math. Ann. 77 (3) (1916) 313–352.